

Machine Learning Approaches for Early Detection of Neurodegenerative Disorders Using Multimodal Biomarker Data

Internal Seed Grant Proposal

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Project Summary. Neurodegenerative disorders including Alzheimer's and Parkinson's disease affect over 50 million people worldwide, yet early detection remains a critical clinical challenge. This proposal seeks \$48,750 in seed funding to develop a machine learning pipeline that integrates multimodal biomarker data — neuroimaging, cerebrospinal fluid proteomics, and wearable sensor time series — for early risk stratification. The 12-month project leverages existing clinical datasets from the Meridian University Hospital system ($n \approx 1,200$ patients) and will establish preliminary data and computational infrastructure for a competitive NIH R01 submission in 2028. The proposed architecture combines transformer-based sequence models with cross-modal attention mechanisms to exploit complementary biomarker signals. Anticipated outcomes include a validated predictive model, a curated multimodal benchmark dataset, two peer-reviewed publications, and a fully drafted R01 proposal.

Research Plan & Objectives

Neurodegenerative diseases impose a growing societal and economic burden. Current diagnostic criteria for Alzheimer's and Parkinson's rely predominantly on clinical symptom presentation, which emerges only after substantial irreversible neural loss has occurred. While individual biomarker modalities — structural MRI, amyloid-PET, CSF protein assays, and actigraphy-derived motor assessments — each carry predictive signal, no existing computational framework systematically integrates these heterogeneous data streams for early, pre-symptomatic risk assessment.

The central hypothesis is that a multimodal transformer architecture with learned cross-modal attention can identify subtle, interacting biomarker signatures preceding clinical diagnosis by 2–5 years, outperforming single-modality baselines and current clinical screening tools.